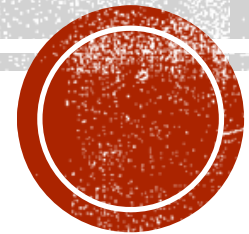


INTRODUCTION TO RECONFIGURABLE COMPUTING

CS G553: Reconfigurable computing

Dr. Meetha V Shenoy

Lect-8



LUT-based Technology Mapping

Algorithms to minimize area

Chortle-crf

MIS-fpga

Xmap

Algorithms delay minimization

FlowMap

*Chortle-
d*

*DAG-
map*

*MIS-fpga-
delay*

Algorithms for maximizing routability



Chortle-crf

The Chortle algorithm was developed by Francis et al. [86, 87] at the University of Toronto in 1991

Aim: Minimizing the amount of LUTs in the implementation

Input: SOP representation of a single output function, K-input LUTs

Bin Packing Algorithm



Chortle-crf

Algorithm operates in two steps:

Step 1: Original Boolean network is partitioned into a forest of trees *that are then separately mapped into circuits of K -input LUTs*.

Step 2: The circuits implementing the trees are assembled to produce the final circuit.



Chortle-crf

Algorithm operates in two steps:

Step 1: Original Boolean network is partitioned into a forest of trees *that are then separately mapped into circuits of K -input LUTs.*



Chortle-crf

Mapping the tree

Combination of bin packing algorithm and dynamic programming

Each tree is traversed from primary inputs to primary outputs

At each node v a circuit referred to as the *best circuit* is constructed.

Best circuit implements the cone at v extending from v to the primary inputs, using least number of LUTs



Chortle-crf

Bin Packing Algorithm

Placing n objects into a number of bins (at most n bins).

Each object has a weight ($W_i > 0$) and Each bin has a limited capacity ($C_i > 0$)



Chortle-crf

Bin Packing Algorithm

Find the best assignment of objects to bins such that:

The total weight of the objects in each bin does not exceed its capacity

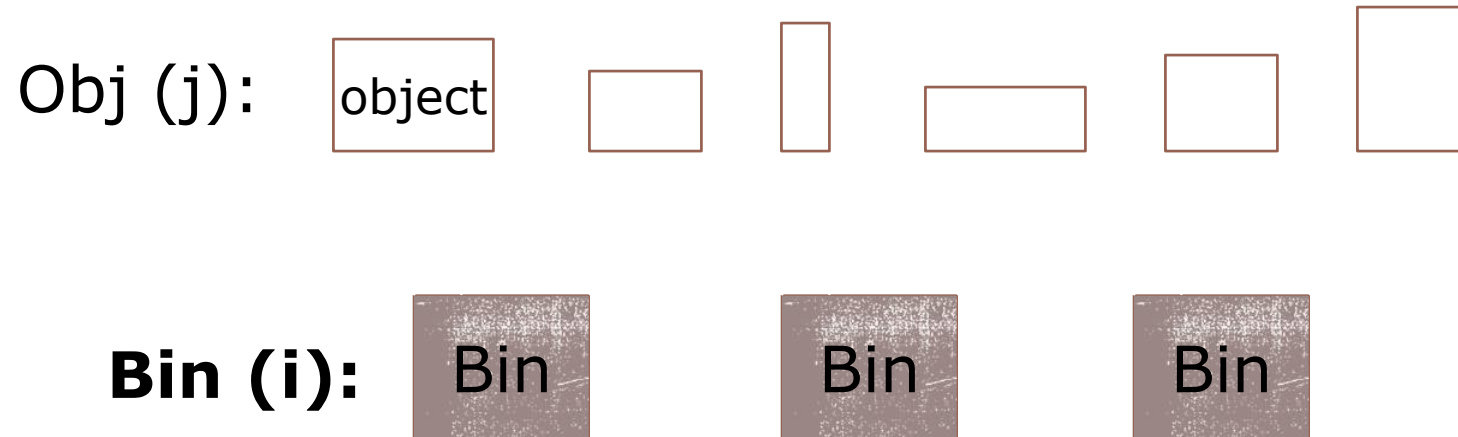
The number of bins used is minimized



Chortle-crf

Bin Packing Algorithm

Let $Y_i = 1$ if (bin i) is used, Let $X_{ij} = 1$ if (object j) is assigned to (bin i)



Bin Packing

$$\begin{aligned} \text{Min } Z &= \sum_{i=1}^n Y_i \\ \text{S.T } \sum_{j=1}^n W_j X_{ij} &\leq C Y_i \quad i \in N = \{1, 2, \dots, n\} \\ \sum_{i=1}^n X_{ij} &= 1 \quad j \in N \end{aligned}$$

Let $Y_i = 1$ if (bin i) is used, Let $X_{ij} = 1$ if (object j) is assigned to (bin i)



Chortle-crf

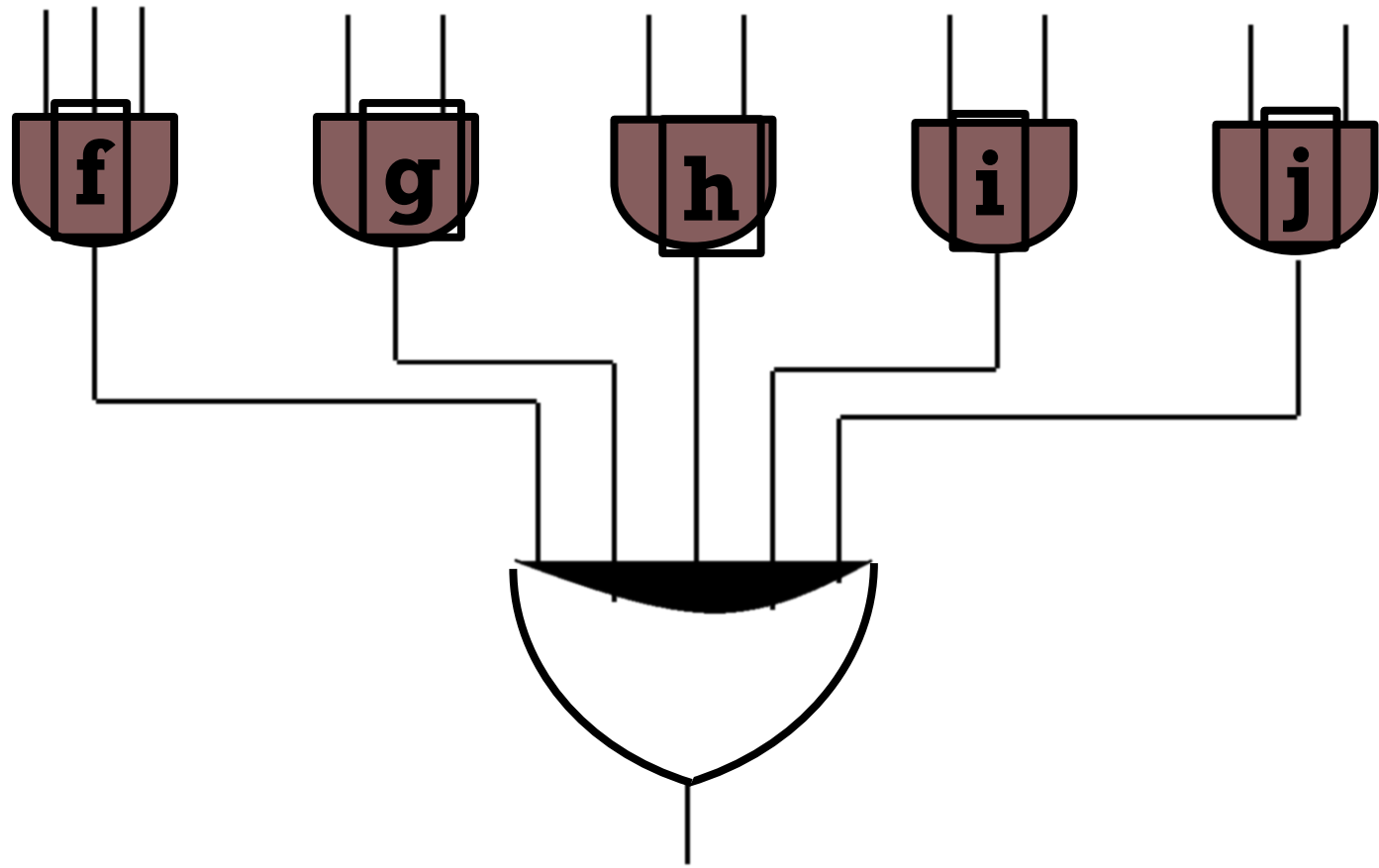
Implementation of Tree

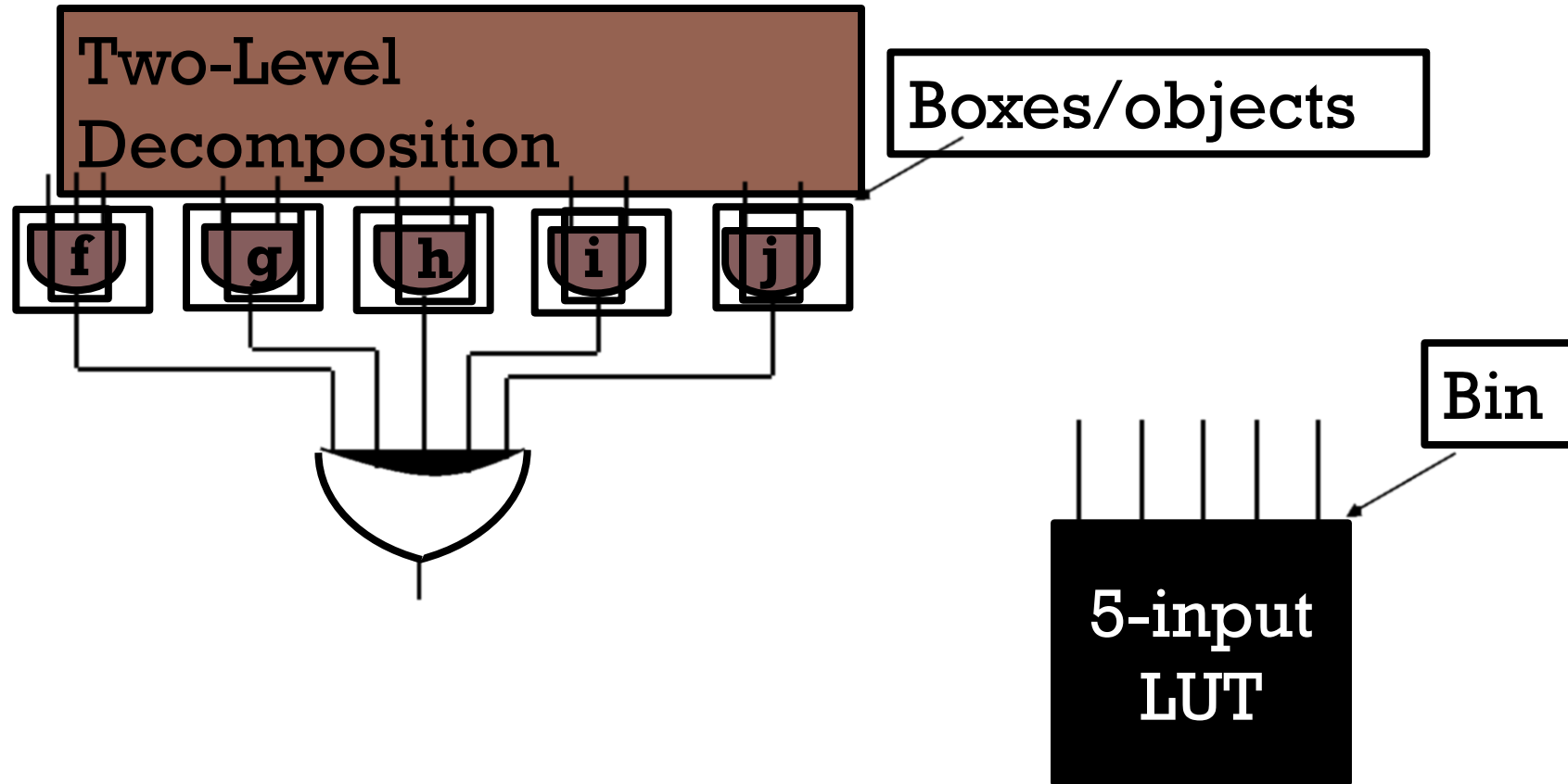
Two-Level Decomposition

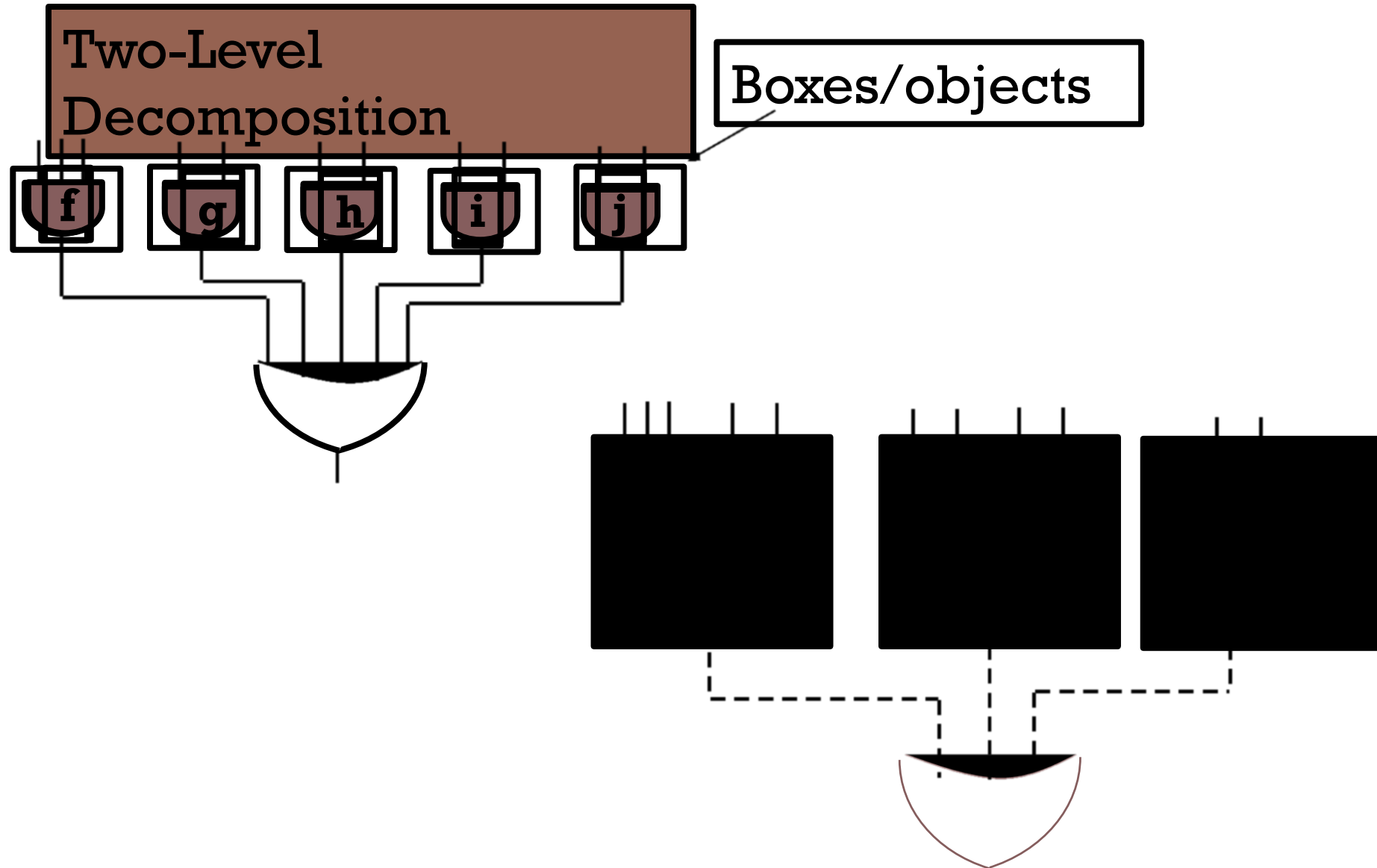
Multi-Level Decomposition



Two-Level Decomposition







Chortle-crf

Two-Level
Decomposition

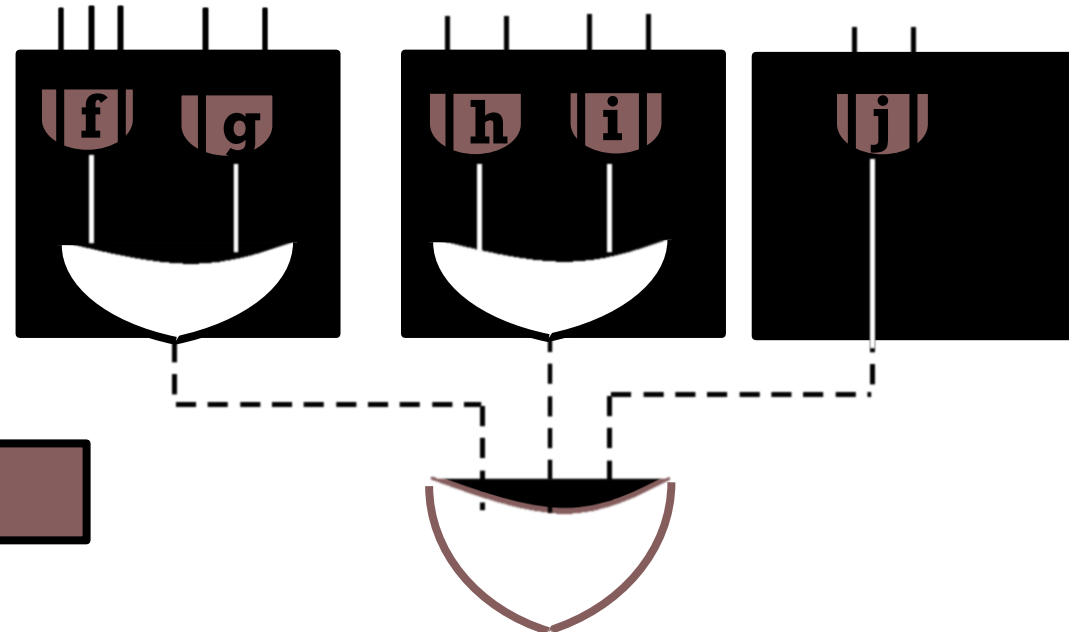
Boxes/objects



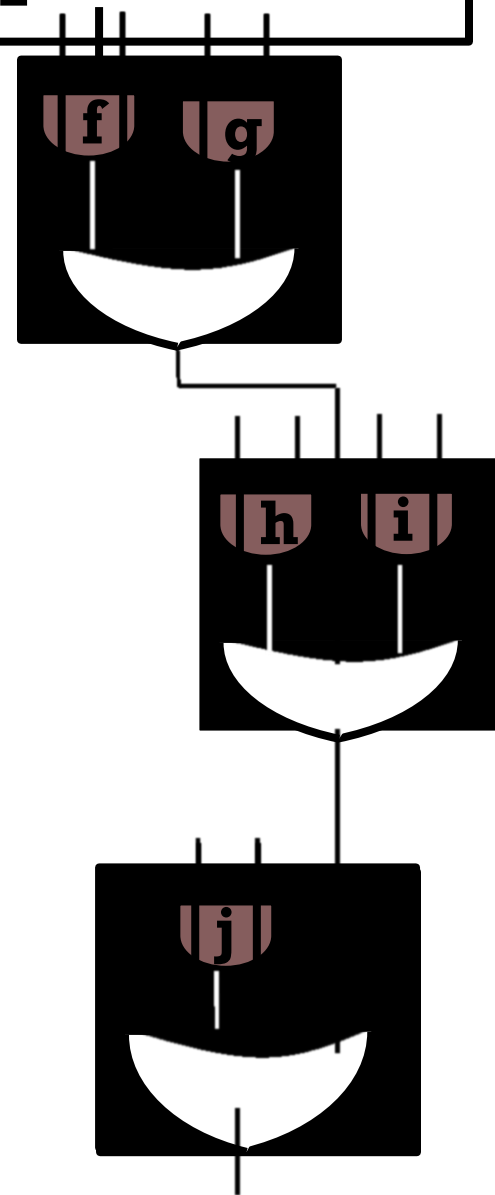
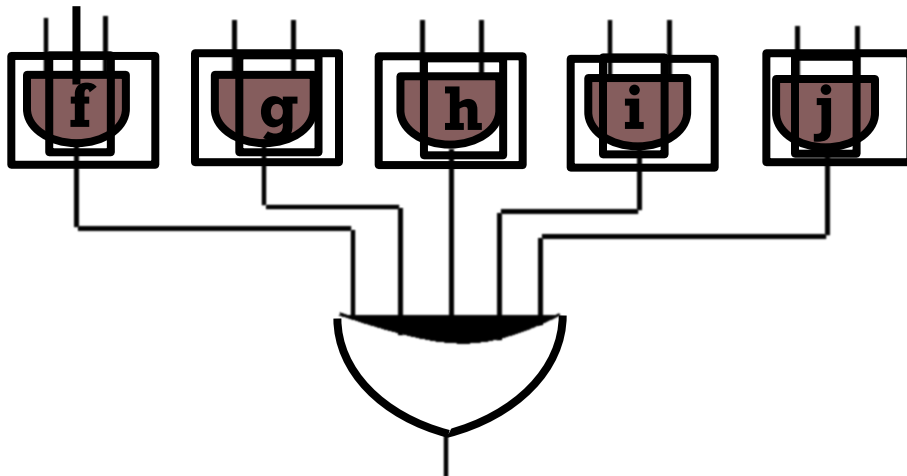
3, 2, 2, 2, 2

5, 4, 2

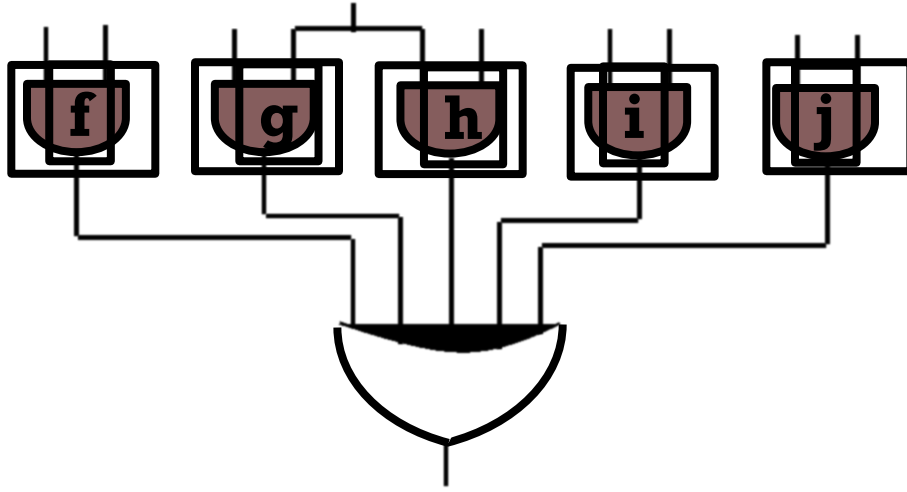
First Fit decreasing method



Chortle-crf

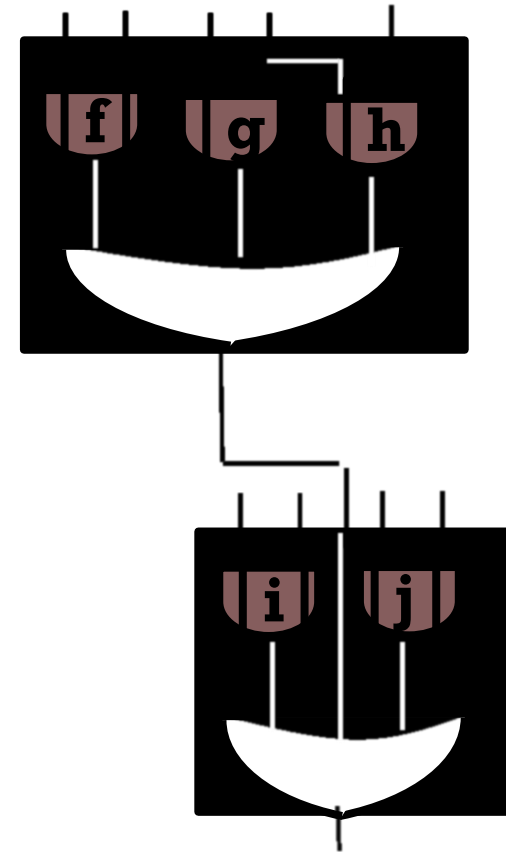
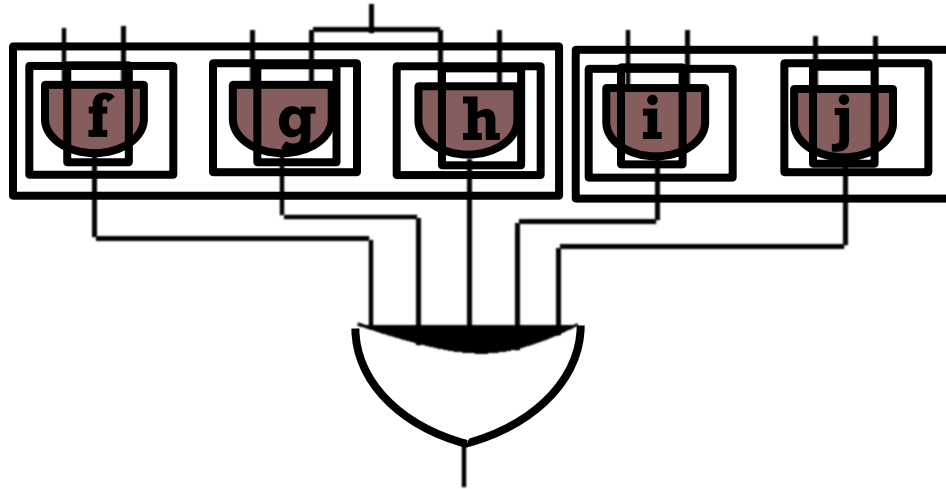


Exploiting Reconvergent Paths



Chortle-crf

Exploiting Reconvergent Paths



Chortle-crf

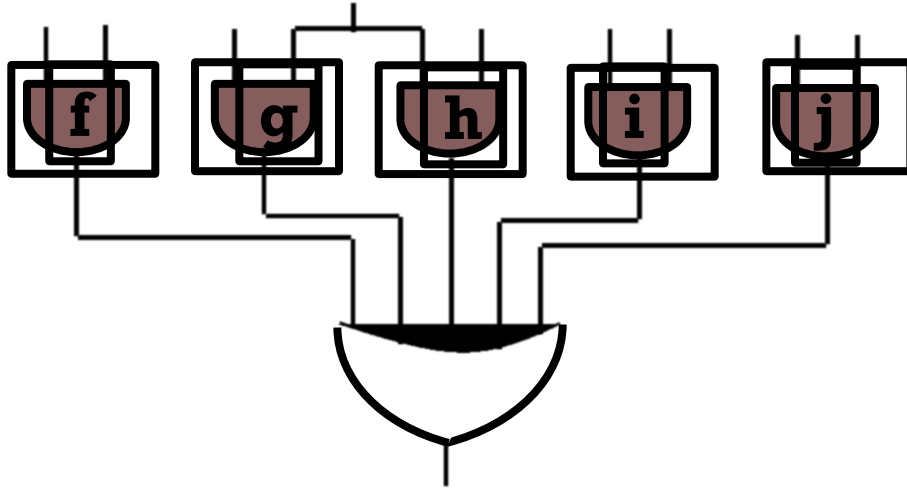
Multi-Level Decomposition

MultiLevel

```
{
while there is more than one unconnected bin
{
    if there are no free inputs among the remaining unconnected bins
    {
        create an empty bin and add it to the end of the list
    }
    connect the most filled unconnected bin to the next unconnected bin with a free
    input
}
}
```

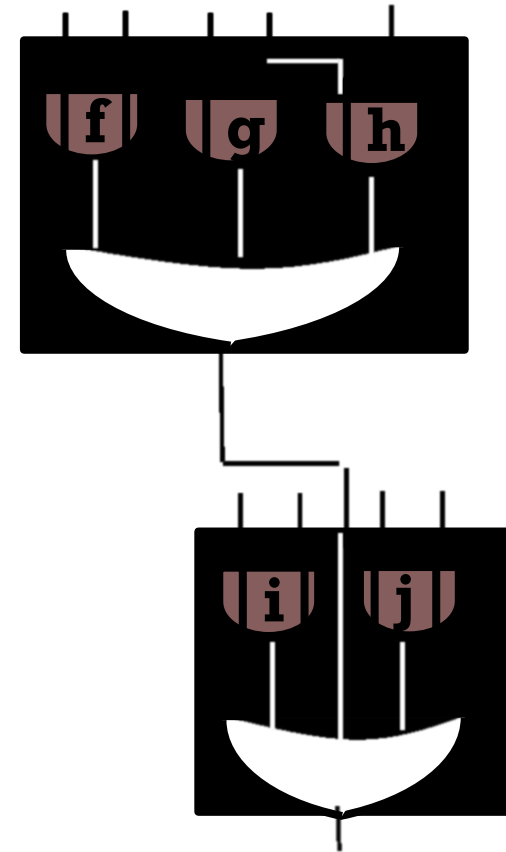
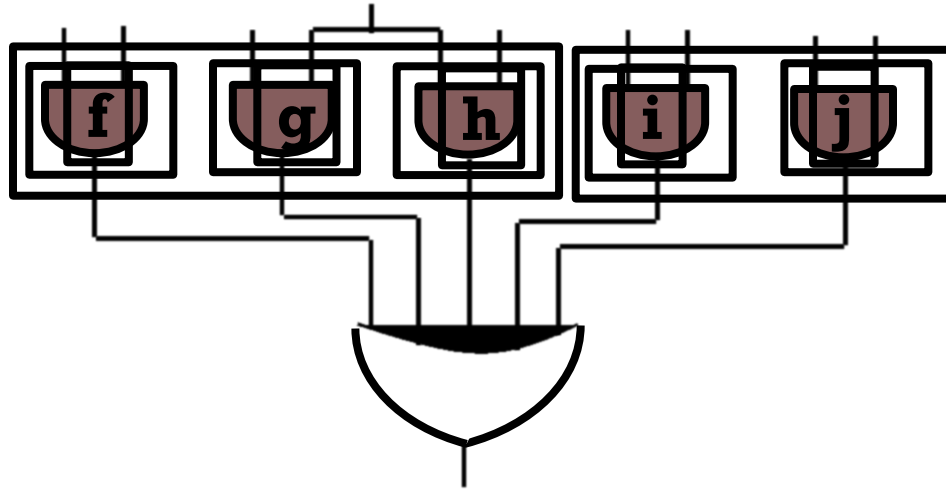


Exploiting Reconvergent Paths

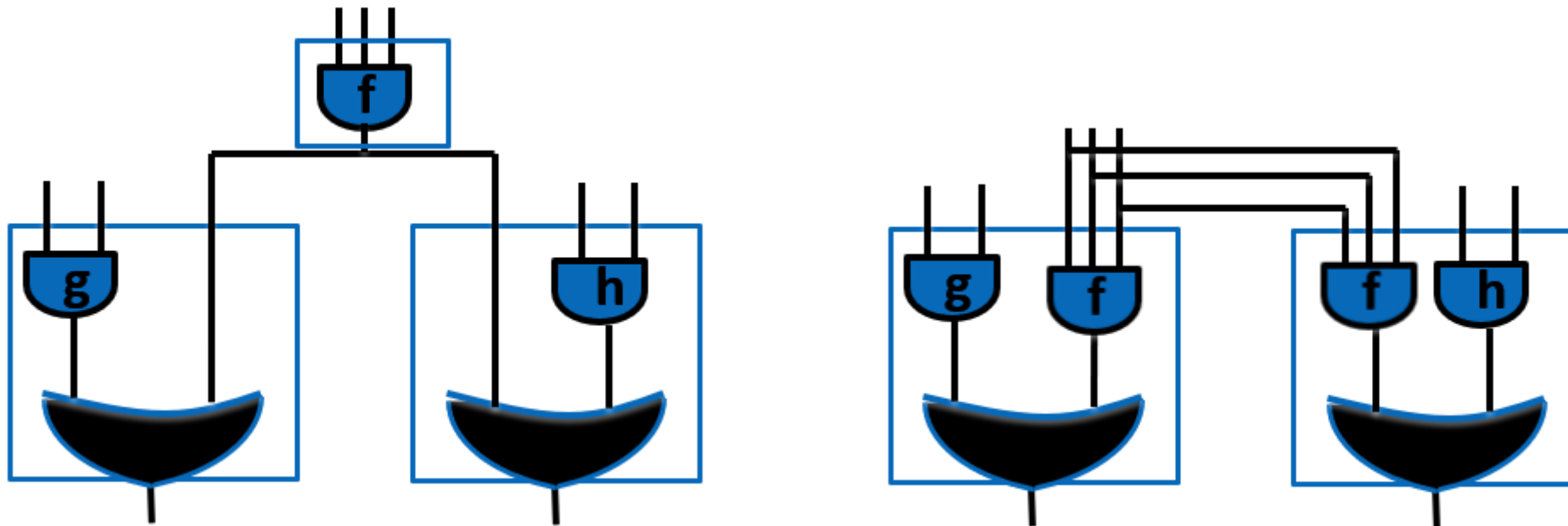


Chortle-crf

Exploiting Reconvergent Paths



Logic Replication



Chortle-crf

- c using only the constructive bin packing approach
- cr using the reconvergent optimization
- cf using the replication optimization
- crf using both reconvergent and replication



Exercises

Apply Chortle-crf LUT-mapping algorithm for the expression below with **K=4**: (show all the steps of the algorithm)

$$Y = ABC + ACE + DB + DFGH$$



LUT-based Technology Mapping

Algorithms to minimize area

Chortle-crf

MIS-fpga

Xmap

Algorithms delay minimization

FlowMap

*Chortle-
d*

*DAG-
map*

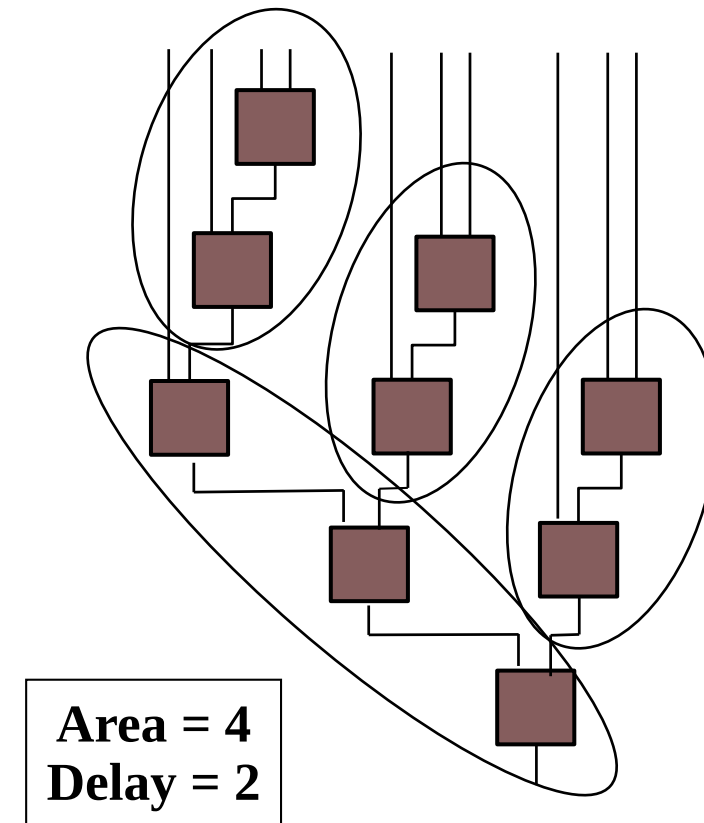
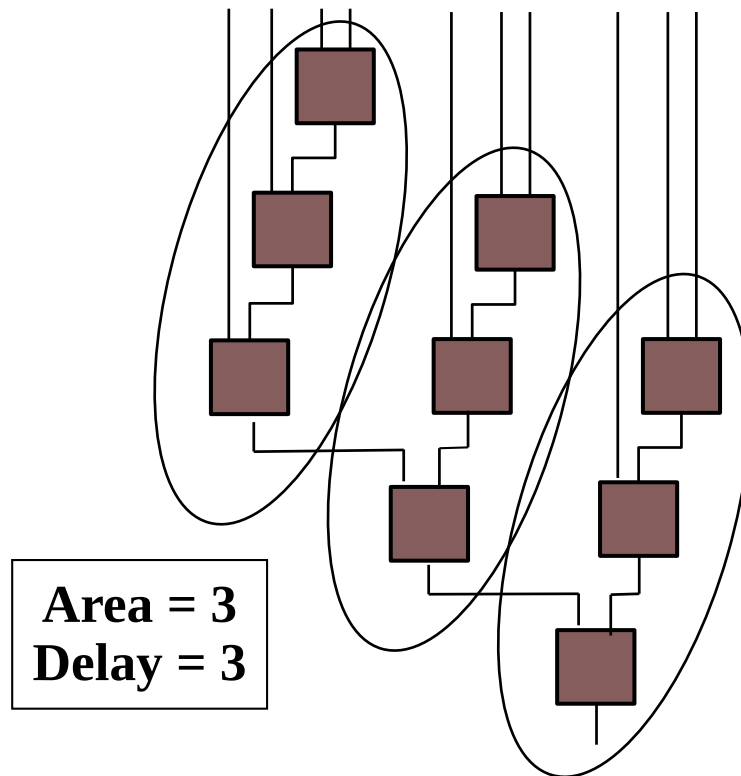
*MIS-fpga-
delay*

Algorithms for maximizing routability



FLOW MAP-INTRO

- Optimizing for area vs. optimizing for delay



FLOWMAP-INTRO

Objective:

- o Minimizing **signal delays** of mapped designs

Signal delay:

- o Delay in the LUTs
- o Interconnection delay



FLOW MAP

- LUT placement is not known
 - Only LUT delay is considered
 - Interconnection delay : assumed to be the same for all signals
 - The delay of a signal = the number of LUTs that the signal traverses on a path from input to output
 - minimization of the depth of the resulting DAG



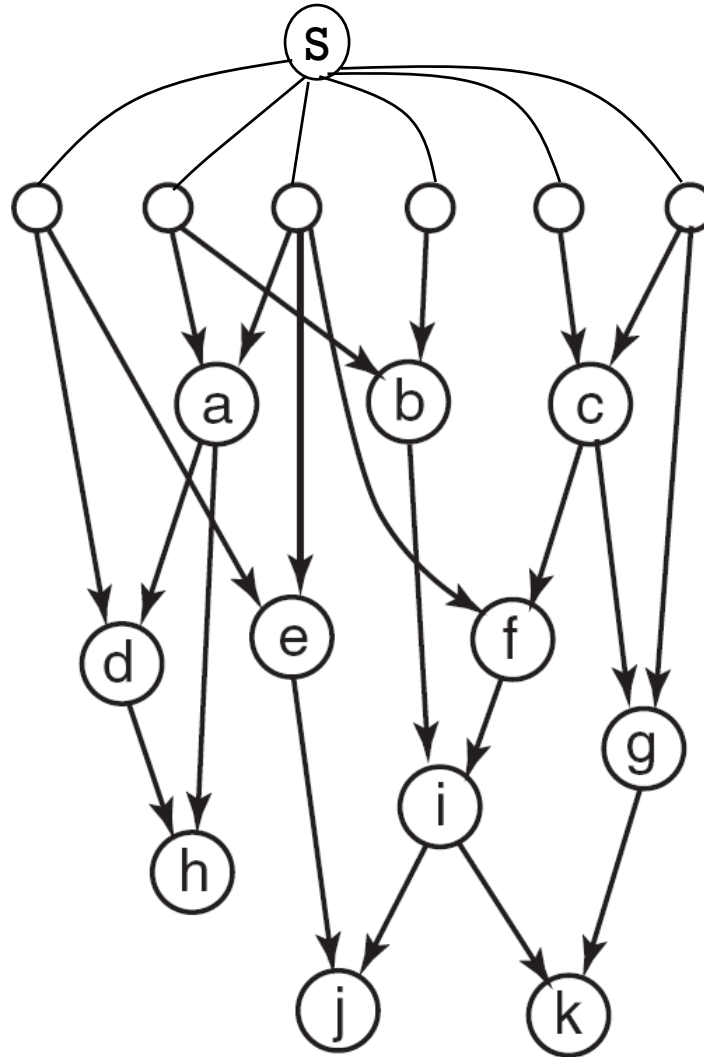
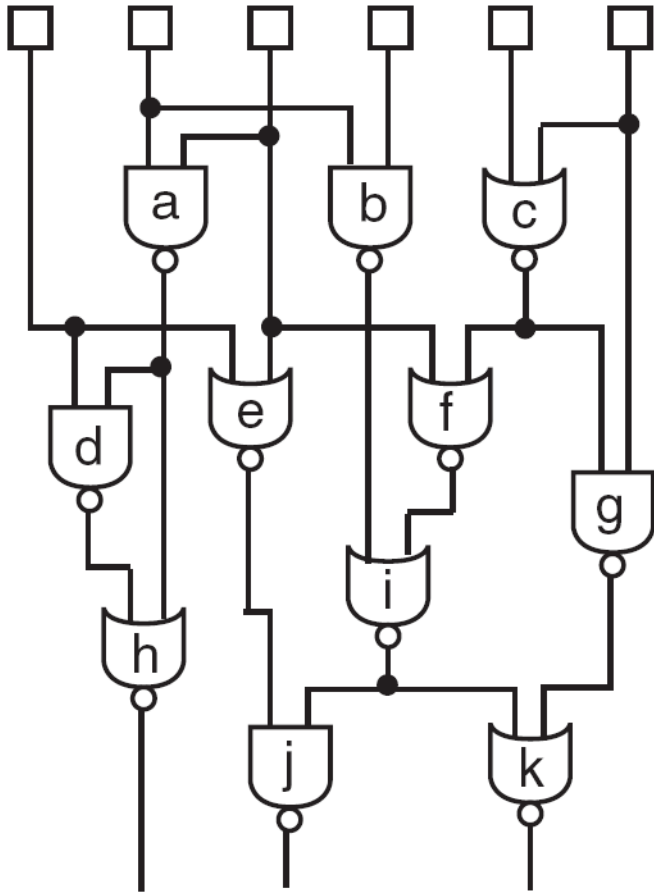
FLOW MAP

- Two Steps:
 - Node labelling
 - Node mapping



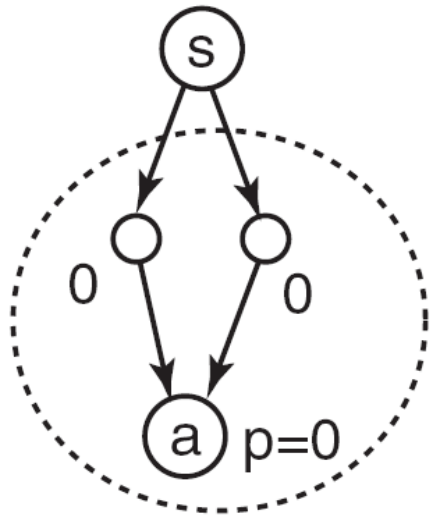
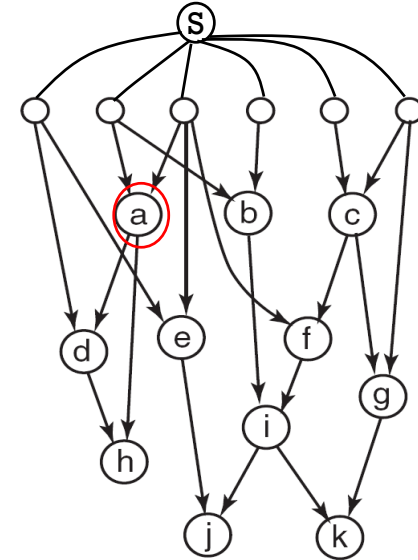
Node Labelling

Example: $K=3$



$l(i) = 0$ for all PIs

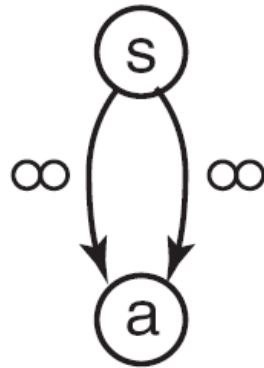
Topological order: $\{a, b, c, d, e, f, g, h, i, j, k\}$



N_a



N'_a



N''_a

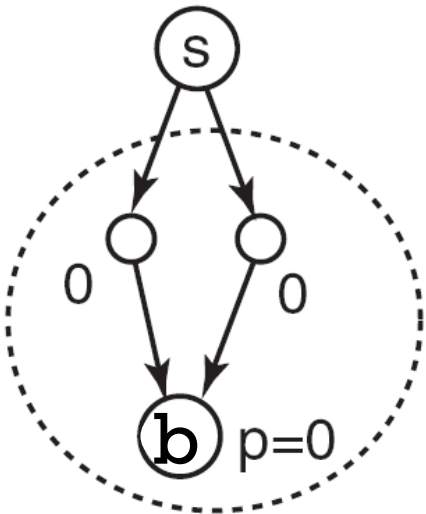
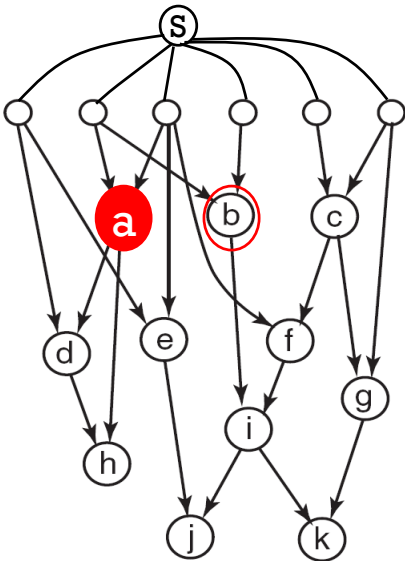
$\overline{X_a} = \{a\}$

$l(a) = 1$

No K-feasible Cut



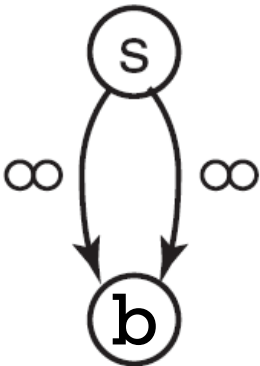
Topological order: {b, c, d, e, f, g, h, i, j, k}



N_b



N'_b



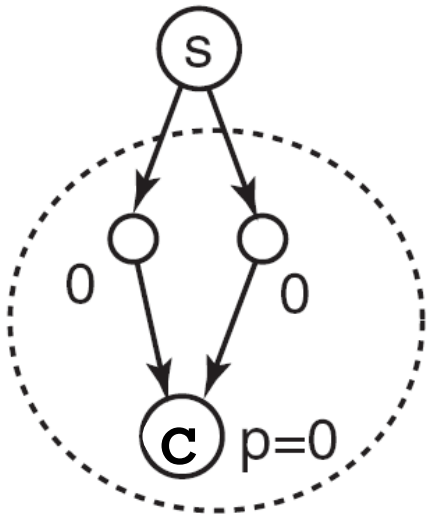
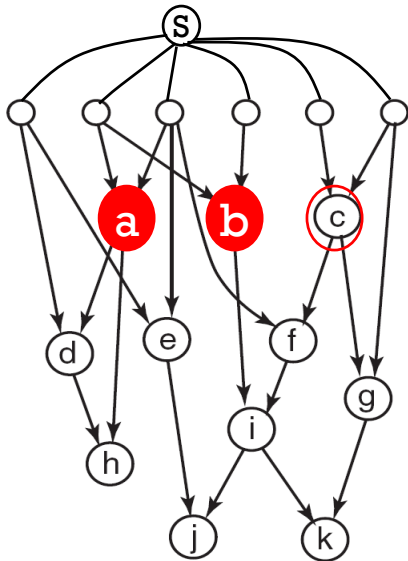
N''_b

$\overline{X_b} = \{b\}$
 $l(b) = 1$

No K-feasible Cut



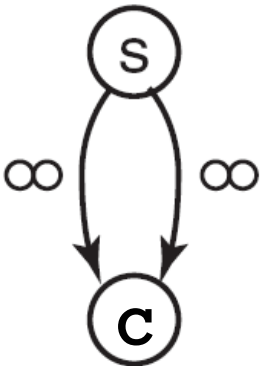
Topological order: {c, d, e, f, g, h, i, j, k}



N_c



N'_c



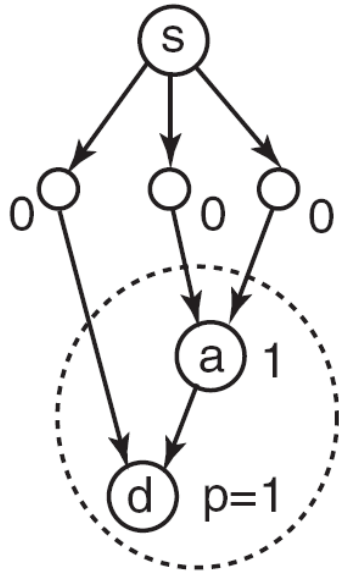
N''_c

$\overline{X_c} = \{c\}$
 $l(c) = 1$

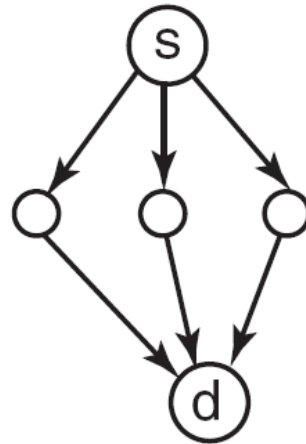
No K-feasible Cut



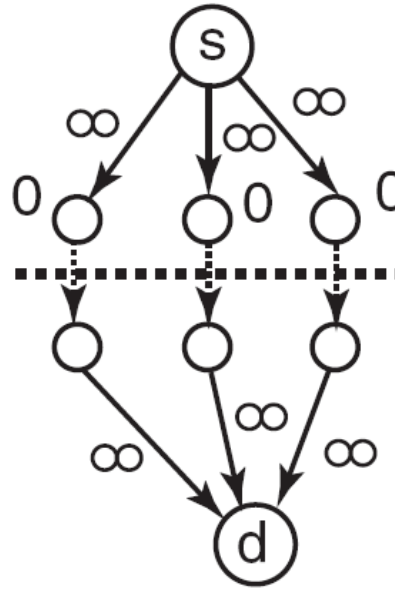
Topological order: {d, e, f, g, h, i, j, k}



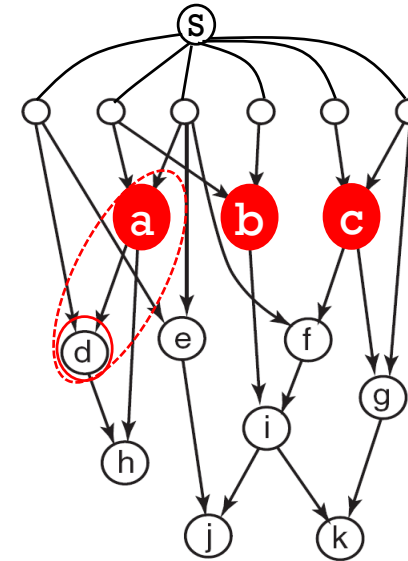
N_d



N'_d



N''_d

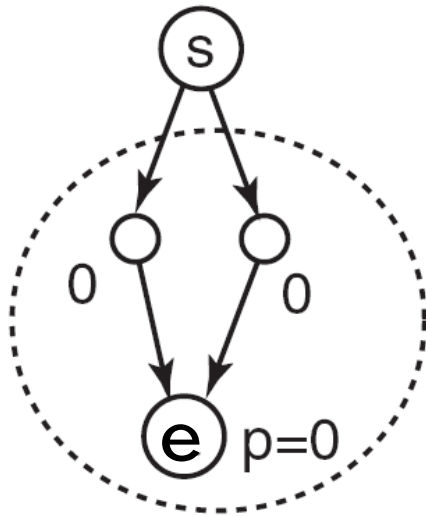


$$\overline{X_d} = \{a, d\}$$

$$l(d) = 1$$



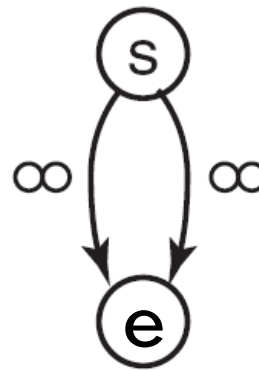
Topological order: {e, f, g, h, i, j, k}



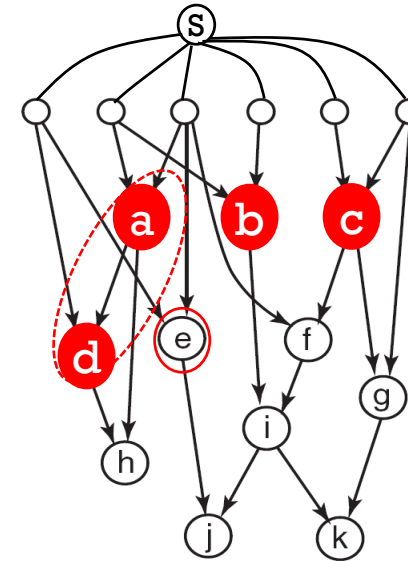
N_e



N'_e



N''_e



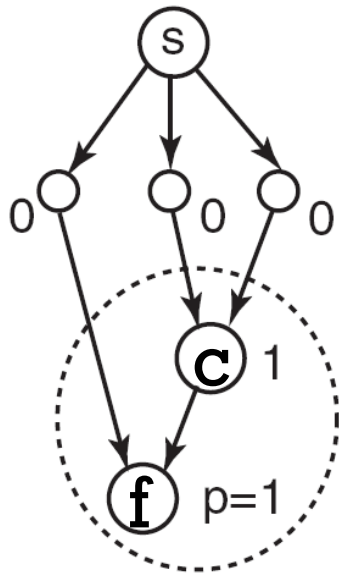
$$\overline{X_e} = \{e\}$$

$$l(e) = 1$$

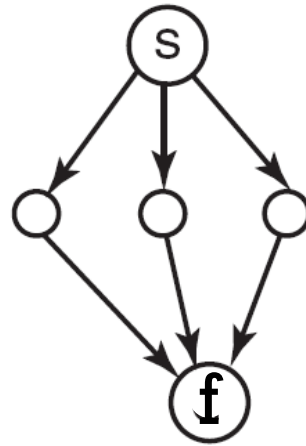
No K-feasible Cut



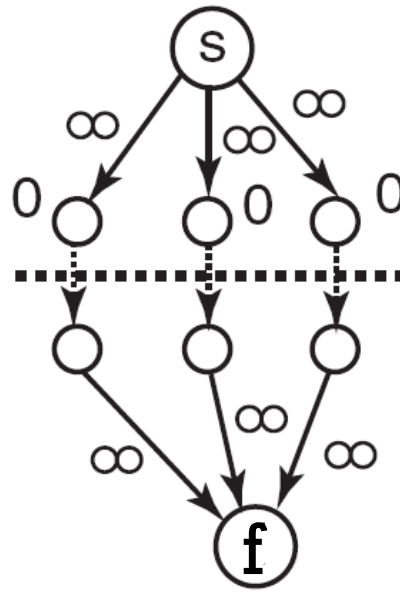
Topological order: {f, g, h, i, j, k}



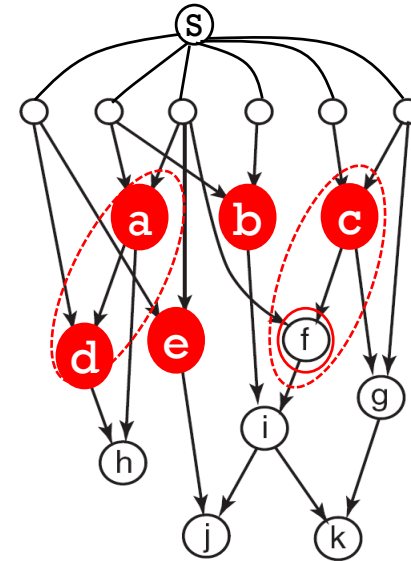
N_f



N'_f



N''_f

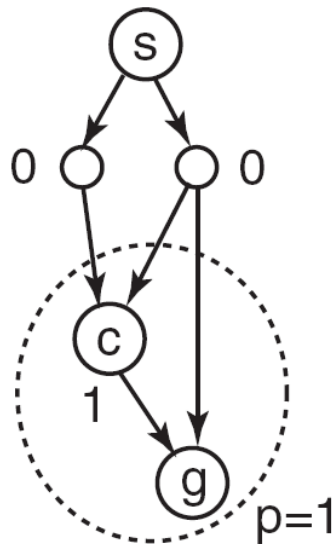


$$\overline{X}_f = \{c, f\}$$

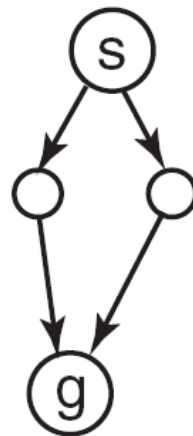
$$l(f) = 1$$



Topological order: {g, h, i, j, k}

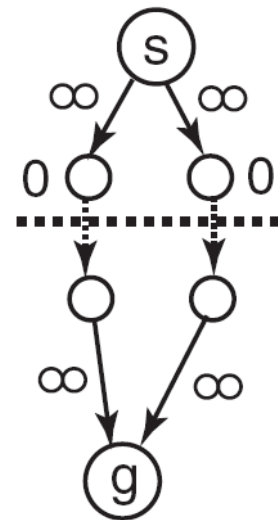


N_g

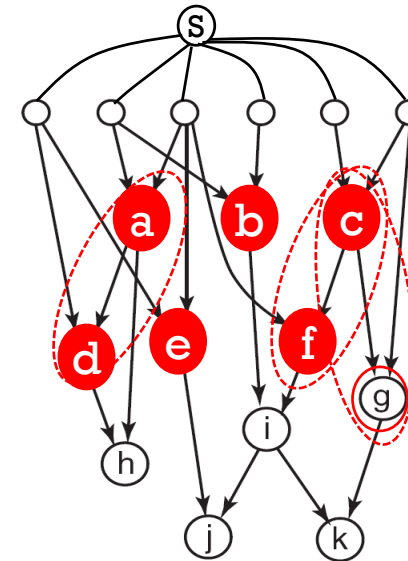


N'

g



N''_g

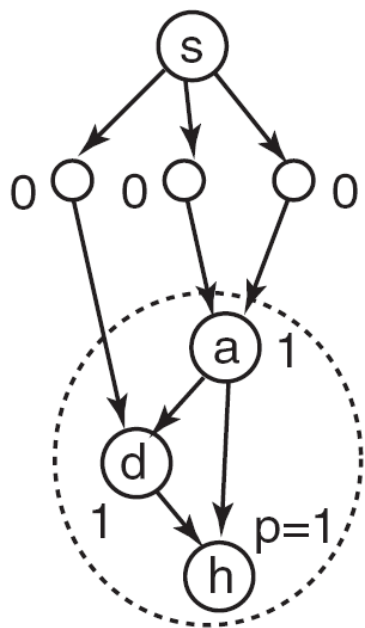


$$\overline{X_g} = \{c, g\}$$

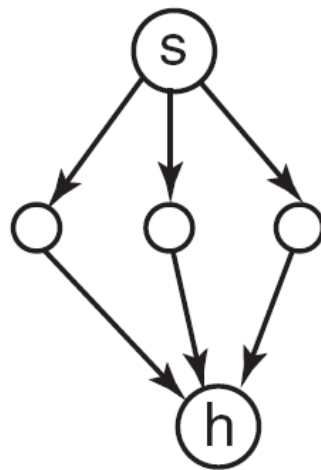
$$l(g) = 1$$



Topological order: {h, i, j, k}

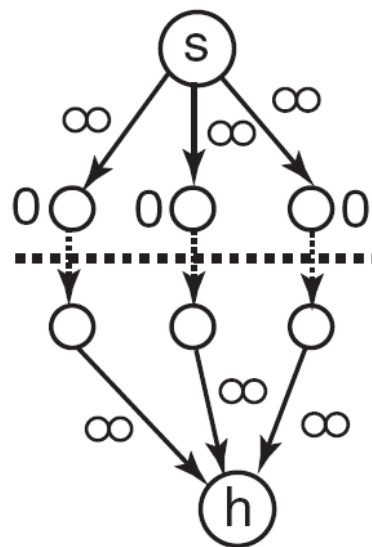


N_h

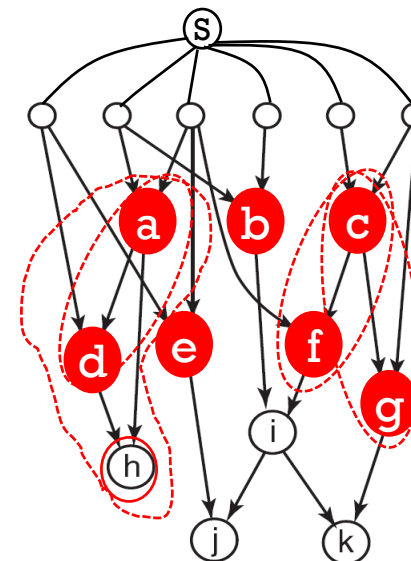


N'

h



N''_h

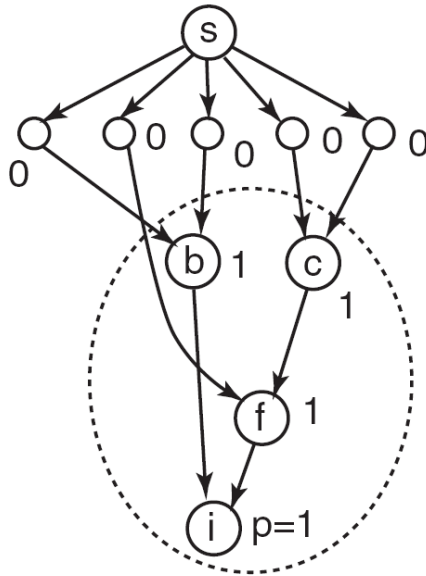


$$\overline{X_h} = \{a, d, h\}$$

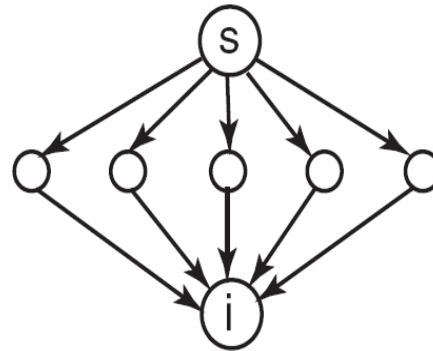
$$l(h) = 1$$



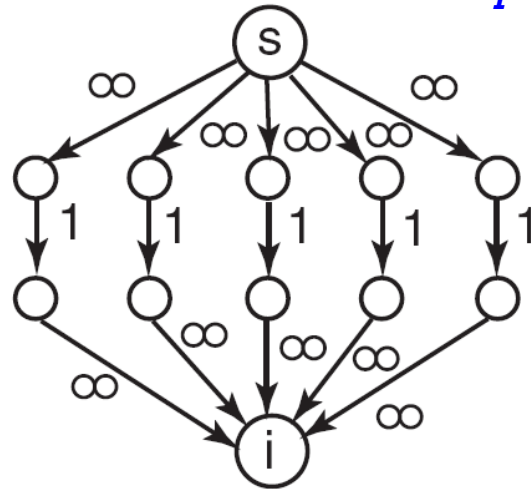
Topological order: $\{i, j, k\}$



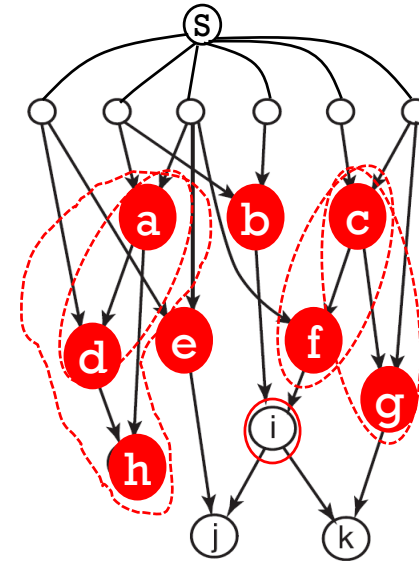
N_i



N'_i



N''_i



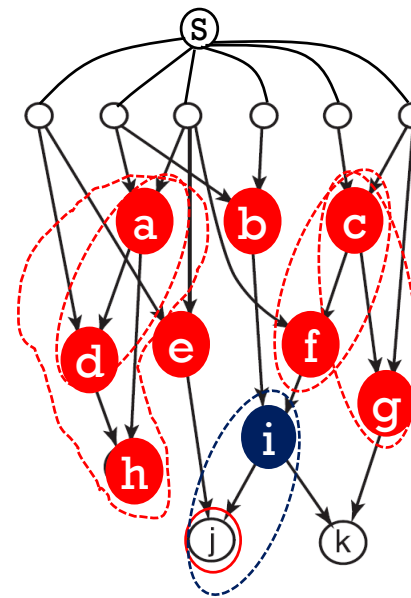
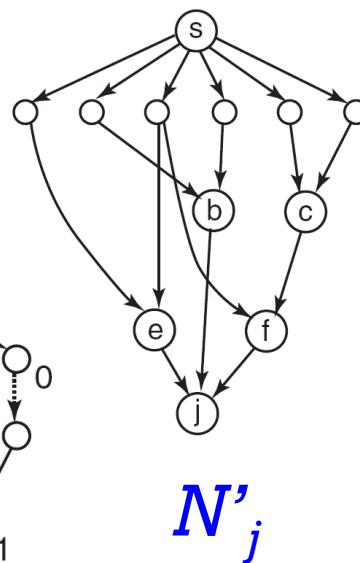
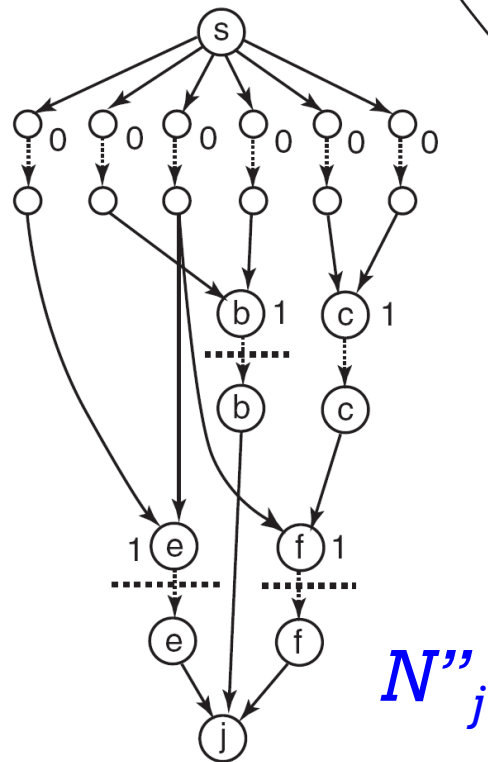
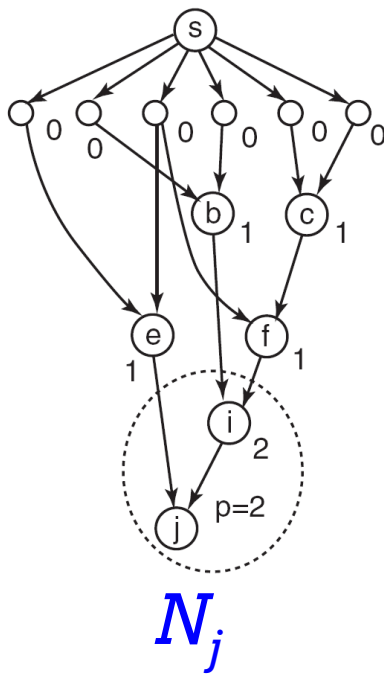
$\bar{X}_i = \{i\}$

$l(i) = 2$

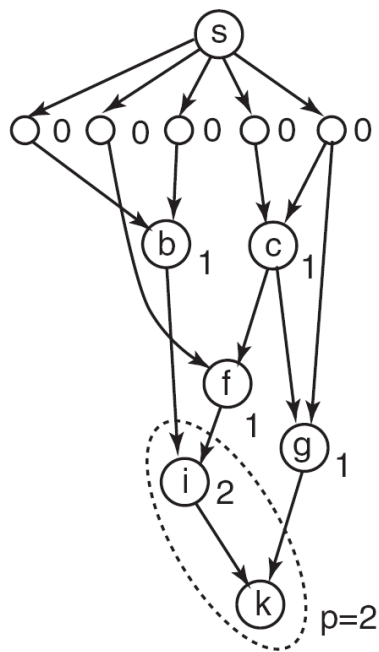
No K-feasible Cut



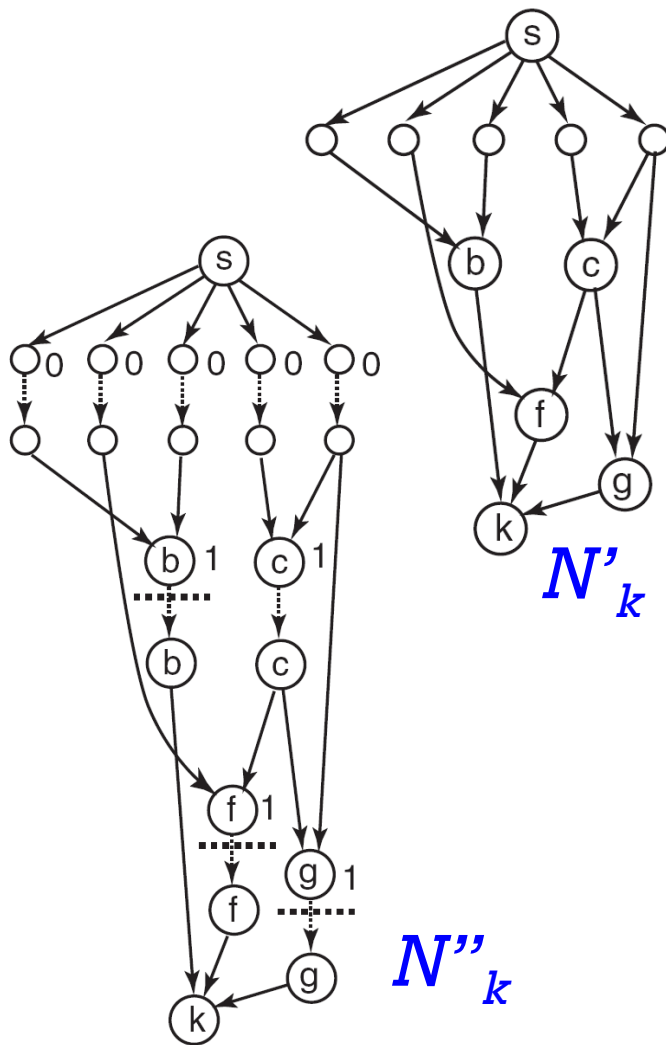
Topological order: $\{j, k\}$



Topological order: {k}

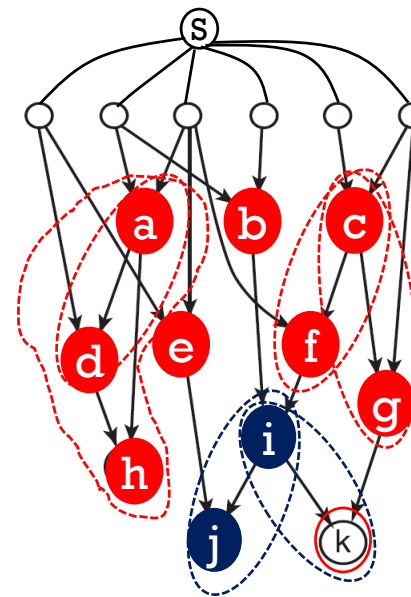


N_k



N'_k

N''_k



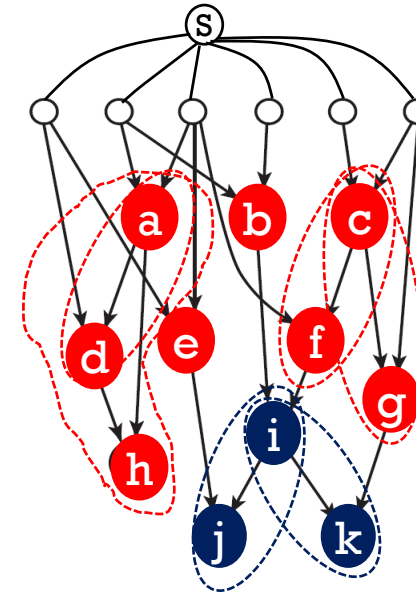
$$\bar{X}_j = \{i, k\}$$

$$l(k) = 2$$



Topological order: {}

Node	Label	Clustering
a	1	$\{a\}$
b	1	$\{b\}$
c	1	$\{c\}$
d	1	$\{a, d\}$
e	1	$\{e\}$
f	1	$\{c, f\}$
g	1	$\{c, g\}$
h	1	$\{a, d, h\}$
i	2	$\{i\}$
j	2	$\{i, j\}$
k	2	$\{i, k\}$



FlowMap

Node Labelling

Labels every node in topological order

Each Node is processed after all its predecessors

Starting from PI nodes, compute node labels in topological order

Compute the label of a node based on labels of its predecessors

Node Mapping

Map the nodes to LUTs (K-Feasible cones) base on Labels

